

Electromagnetism

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1	State when a wire has a magnetic field around it.	When it carries a current.	☐
2 ★	Describe the magnetic field around a straight current-carrying wire.	Concentric circles around the wire.	☐
3	State what the strength of the magnetic field around a wire depends on.	The size of the current and the distance from the wire	☐
4	What is a solenoid?	A long coil of wire.	☐
5 ★	Describe the magnetic field around a solenoid.	Similar to that of a bar magnet.	☐
6	State where the magnetic field of a solenoid is strong and uniform.	Inside the solenoid.	☐
7	Describe how to make an electromagnet.	Wrap a coil of wire around an iron core and pass a current through it.	☐
8 ★	Give two advantages of an electromagnet over a permanent magnet.	It can be switched on and off, and its strength can be varied.	☐
9 ★	Give two ways to make an electromagnet stronger.	Increase the current or add more turns.	☐
10	State the effect of adding an iron core to a solenoid.	It makes the magnetic field stronger.	☐
11	State how to reverse the magnetic field around a wire.	Reverse the current.	☐
12	A wire is coiled into a solenoid. State the effect on its magnetic field.	It makes the magnetic field stronger.	☐